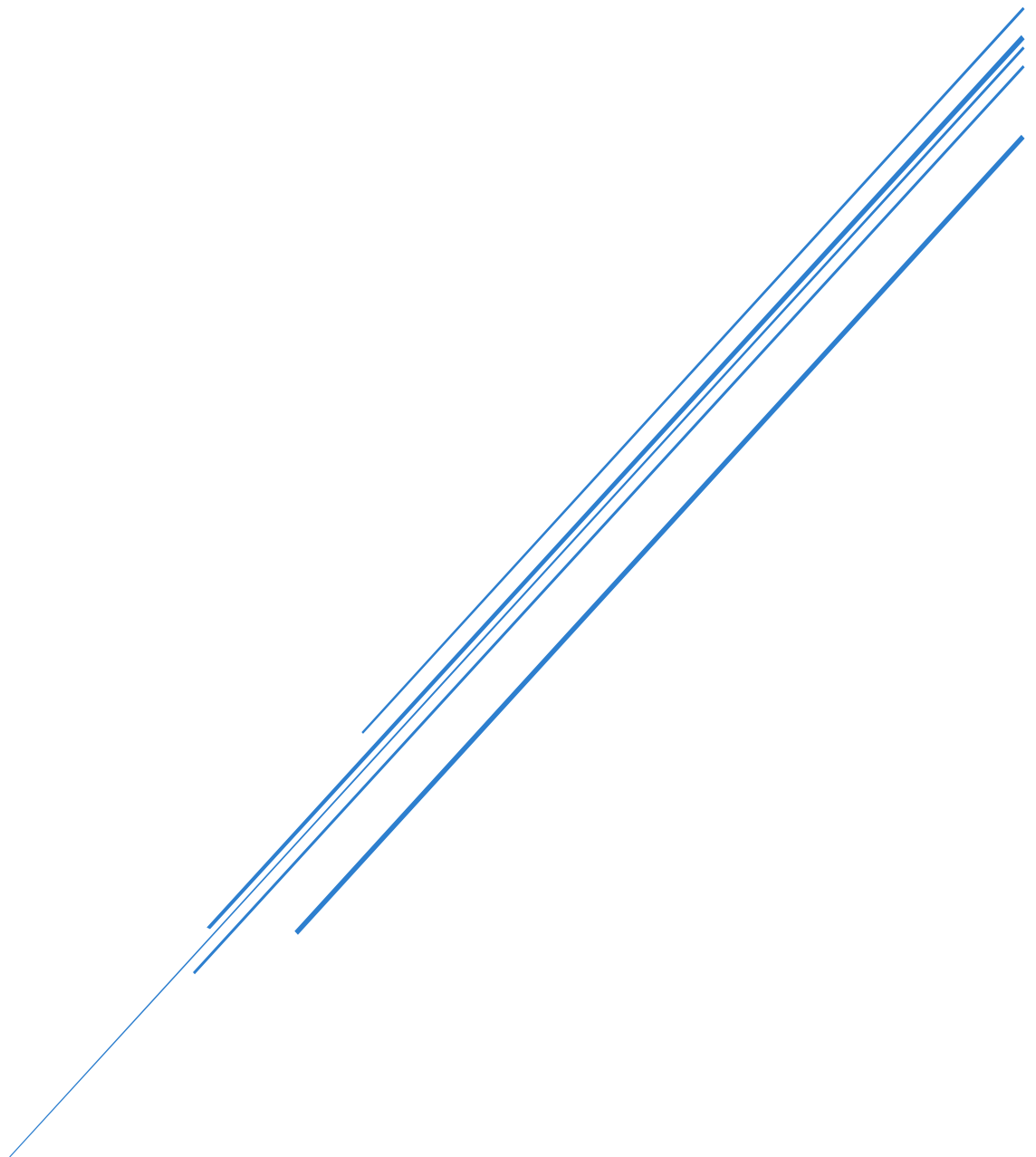


MILESTONE 2

Buffer Overflow Demonstration



Diploma in Information Technology_EHA361
Tshiamo Mosweu_601286

TABLE OF CONTENT

Contents

TABLE OF CONTENT	1
What is a Buffer Overflow?	2
Why Buffer Overflows Matter in Cybersecurity	2
Setup	2
Coding.....	3
Compilation.....	4
Execution	4
Challenges	4
References	5

What is a Buffer Overflow?

A Buffer Overflow happens when a program writes more data into a memory buffer than it was designed to hold (CERT, 2023). Buffers are small, temporary storage areas in memory used to handle data. If the data exceeds the buffer's capacity, the extra information "spills over" into neighbouring memory locations (Stallings, 2018).

This overflow can:

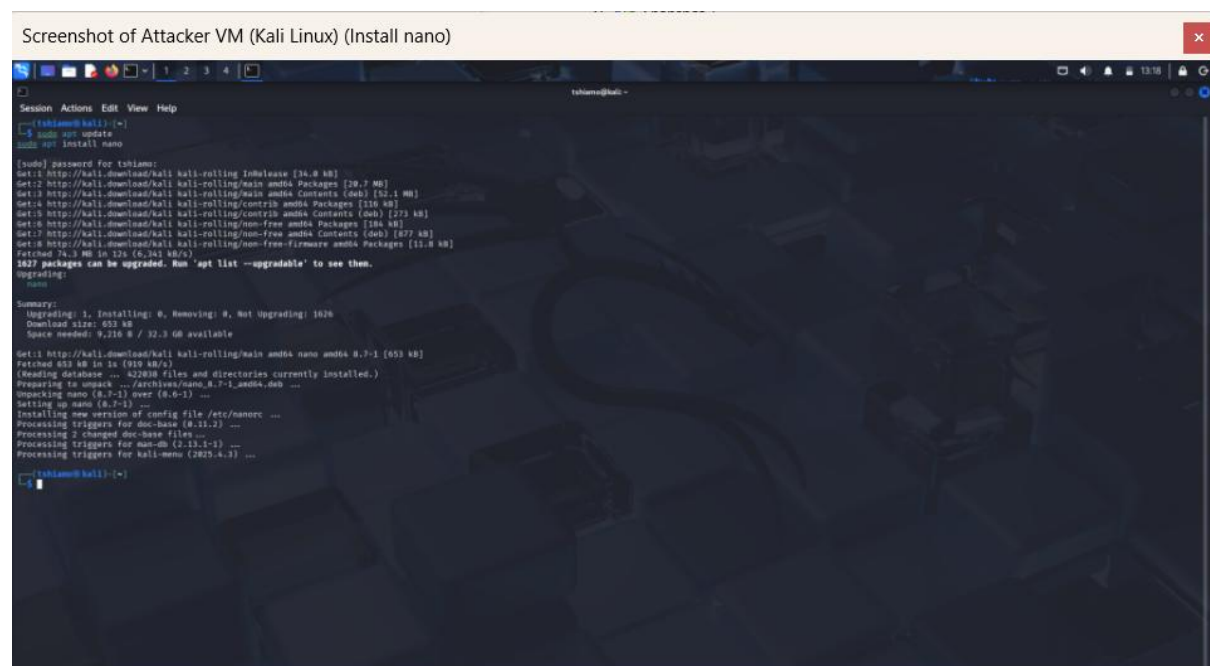
- Overwrite valid data or instructions.
- Cause unpredictable behaviour such as crashes (segmentation faults).
- Create serious security vulnerabilities, since attackers can exploit the overflow to inject malicious code or gain unauthorized access.

Why Buffer Overflows Matter in Cybersecurity

- They are one of the oldest and most common vulnerabilities in software (Aleph One, 1996).
- Many famous exploits relied on buffer overflows, such as those described in Smashing the Stack for Fun and Profit (Aleph One, 1996).
- Modern secure coding practices emphasize bounds checking and safer functions to prevent them (CERT, 2023).

Setup

I installed nano to edit the code after realizing it wasn't pre-installed.



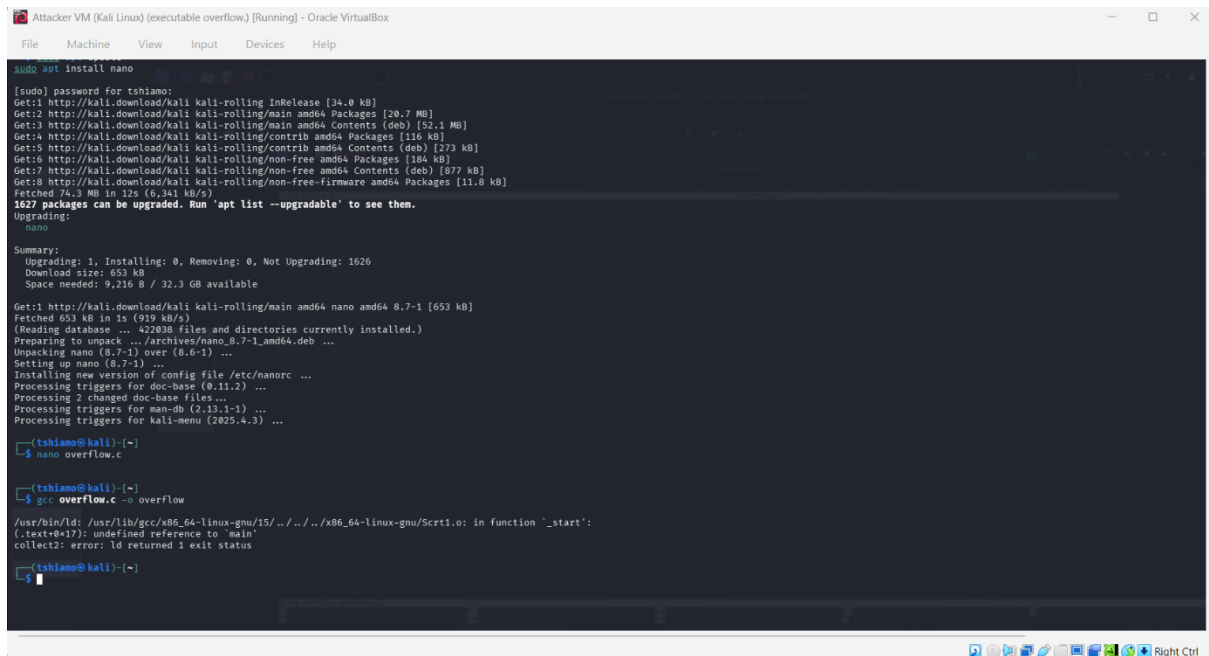
```
Session Actions Edit View Help
--(tshlamm@kali)-[~]
_ $ sudo apt update
sudo apt install nano

[Sube] Password For tshlamm:
Get:1 http://kali.download/kali kali-rolling InRelease [34.8 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [28.7 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [52.1 MB]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [110 kB]
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [278 kB]
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [184 kB]
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [877 kB]
Get:8 http://kali.download/kali kali-rolling/non-free-firmware amd64 Packages [11.8 kB]
Fetched 24.3 MB in 12s (6.34 kB/s)
1627 packages can be upgraded. Run 'apt list --upgradable' to see them.
Upgrading:
nano
Summary:
Upgrading: 1, Installing: 0, Removing: 0, Not Upgrading: 1626
Download size: 933 kB
Space needed: 9,216 B / 32.3 GB available
Get:1 http://kali.download/kali kali-rolling/main amd64 nano amd64 8.7-1 [553 kB]
Fetched 553 kB in 1s (510 kB/s)
(Reading database ... 82808 files and directories currently installed.)
Preparing to unpack .../archives/nano.8.7-1_amd64.deb ...
Unpacking nano (8.7-1) over (8.6-1) ...
Setting up nano (8.7-1) ...
Installing new version of config file /etc/nanorc ...
Processing triggers for doc-base (8.11.2) ...
Processing 2 changed doc-base files ...
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for kali-menu (2023.4.3) ...

--(tshlamm@kali)-[~]
_ $
```

Coding

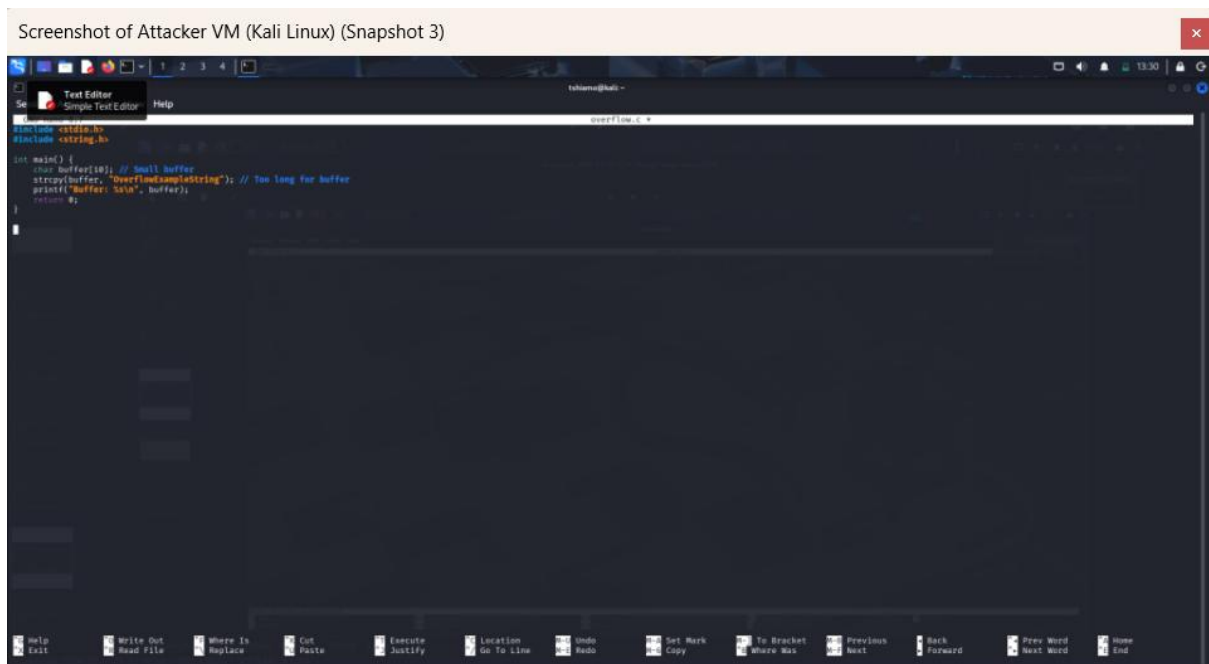
- Initially pasted the code directly into the terminal, which caused errors
- Fixed it by properly saving the program into overflow.c using nano



```
Attacker VM (Kali Linux) (executable overflow) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

[sudo] apt install nano
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [20.7 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [52.1 MB]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [116 kB]
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [273 kB]
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [184 kB]
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [877 kB]
Get:8 http://kali.download/kali kali-rolling/non-free-firmware amd64 Packages [11.8 kB]
Fetched 74.3 MB in 12s (6,341 kB/s)
1627 packages can be upgraded. Run 'apt list --upgradable' to see them.
Upgrading:
nano
Summary:
Upgrading: 1, Installing: 0, Removing: 0, Not Upgrading: 1626
Download size: 653 kB
Space needed: 9,216 B / 32.3 GB available
Get:1 http://kali.download/kali kali-rolling/main amd64 nano amd64 8.7-1 [653 kB]
Fetched 653 kB in 1s (919 kB/s)
(Reading database ... 42208 files and directories currently installed.)
Preparing to unpack .../archives/nano.8.7-1.amd64.deb ...
Unpacking nano (8.7-1) over (8.6-1) ...
Setting up nano (8.7-1) ...
Installing new version of config file /etc/nanorc ...
Processing triggers for doc-base (0.11.2) ...
Processing 2 changed doc-base files...
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for kali-menu (2025.4.3) ...
--(tshiamo@kali)-[-]
└─$ nano overflow.c
--(tshiamo@kali)-[-]
└─$ gcc overflow.c -o overflow
/usr/bin/ld: /usr/lib/gcc/x86_64-linux-gnu/15/../../../../x86_64-linux-gnu/Scrt1.o: in function '_start':
(.text+0x17): undefined reference to 'main'
collect2: error: ld returned 1 exit status
--(tshiamo@kali)-[-]
└─$
```

Coding Section (Error Encountered)



```
Screenshot of Attacker VM (Kali Linux) (Snapshot 3)
Text Editor - Simple Text Editor - Help
overflow.c
#include <stdio.h>
#include <string.h>

int main() {
    char buffer[10]; // Small buffer
    strcpy(buffer, "overflow(overflow)"); // Too long for buffer
    printf("buffer: %s", buffer);
    return 0;
}
```

Compilation

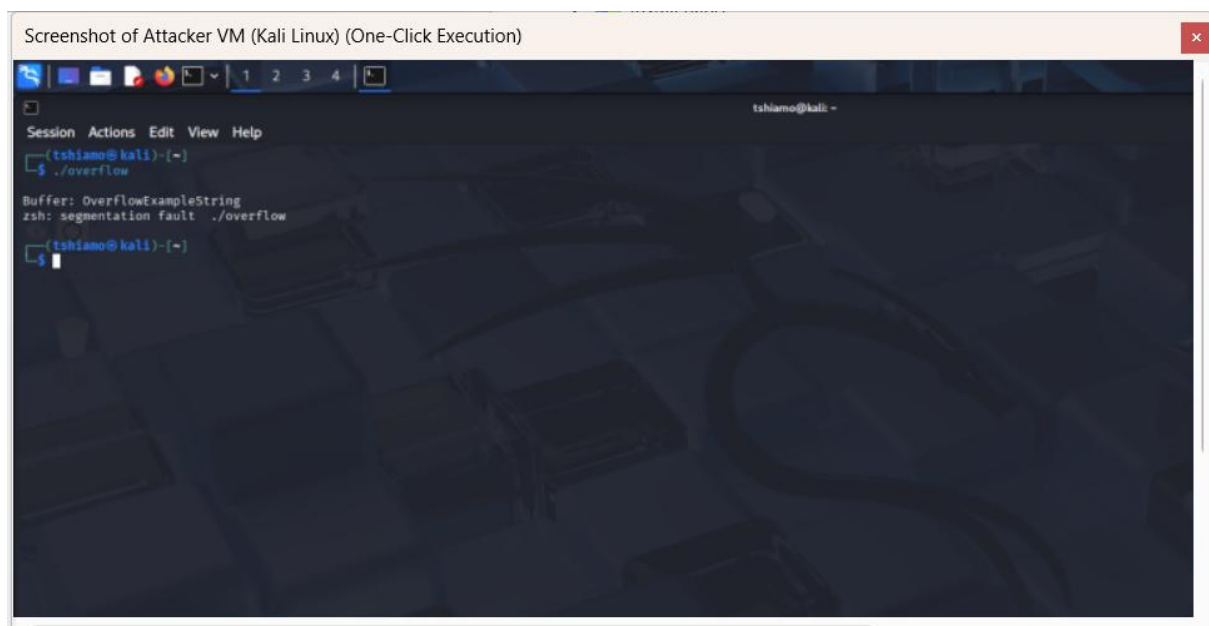
- Compiled with `gcc overflow.c -o overflow`
- During compilation, warnings appeared about writing beyond buffer size. This behaviour is expected in buffer overflow demonstrations, as unsafe functions like `strcpy` do not perform bounds checking (Stallings, 2018).



```
(tshiamo@kali)~$ gcc overflow.c -o overflow
overflow.c: In function 'main':
overflow.c:6:5: warning: 'builtin_memcpy' writing 22 bytes into a region of size 10 overflows the destination [-Wstringop-overflow-]
6 |     strcpy(buffer, "OverflowExampleString"); // Too long for buffer
  |     ~~~~~^~~~~~
overflow.c:5:10: note: destination object 'buffer' of size 10
5 |     char buffer[10]; // Small buffer
  |     ~~~~~
(tshiamo@kali)~$
```

Execution

- Ran the program with `./overflow`
- Output showed the buffer contents, then crashed with a segmentation fault.
- This confirmed the buffer overflow worked.



```
Session Actions Edit View Help
(tshiamo@kali)~$ ./overflow
Buffer: OverflowExampleString
zsh: segmentation fault ./overflow
(tshiamo@kali)~$
```

Challenges

- VM froze during snapshot, causing errors like VBoxSVC unresponsive.
- Fixed by restarting VirtualBox and cleaning up the saved state.
- This showed the importance of stable VM management during experiments.

References

- Aleph One (1996). Smashing the Stack for Fun and Profit. Phrack Magazine.
- CERT (2023). Buffer Overflow Vulnerabilities. Software Engineering Institute, Carnegie Mellon University.
- Stallings, W. (2018). Operating Systems: Internals and Design Principles. Pearson.

